

# Poster



# Sections

## Background

- Recent ethical dilemma (creatives vs technology)
- AI music is especially difficult to identify
- Audio features are used to investigate this similarity, often by Mel-Frequency Cepstral Coefficients (MFCCs) which measure the variations in frequency, corresponds to timbre and genre; beats per minute (BPM) which measure tempo; and chroma which measures tonal consistency
- Little literature comparing AI and human music, doesn't investigate why it's so difficult to differentiate

## Objective

**Investigating the similarity between AI music and human music via extracted audio features.**

## Methods

- Data collected from Suno (AI-generated music) and Spotify (test and control)
- Extracted audio features from AI and human music
- Aggregated data to investigate audio features within the audio type
- Calculated cosine similarity within each audio type to create sampled pairs dataset
- Conducted K-means clustering between each audio type

## [Visualizations](#)

## Results

- AI-AI has the highest similarity
- AI-Human had slightly lower similarity
- Human-human had the lowest similarity

## Discussion

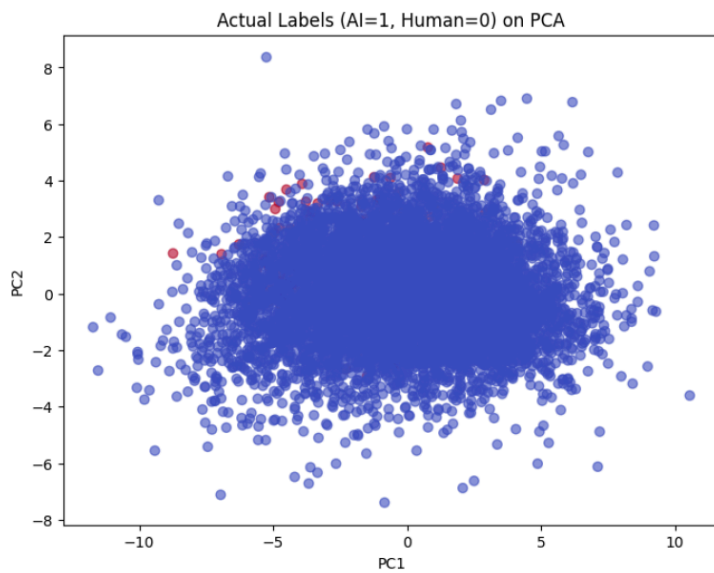
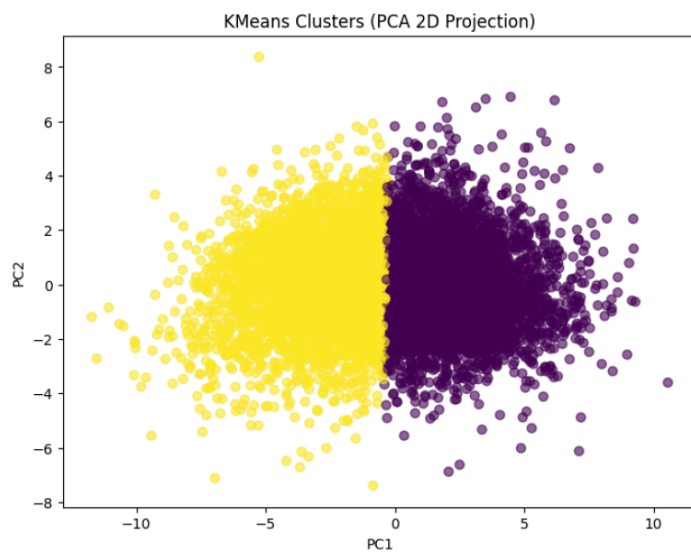
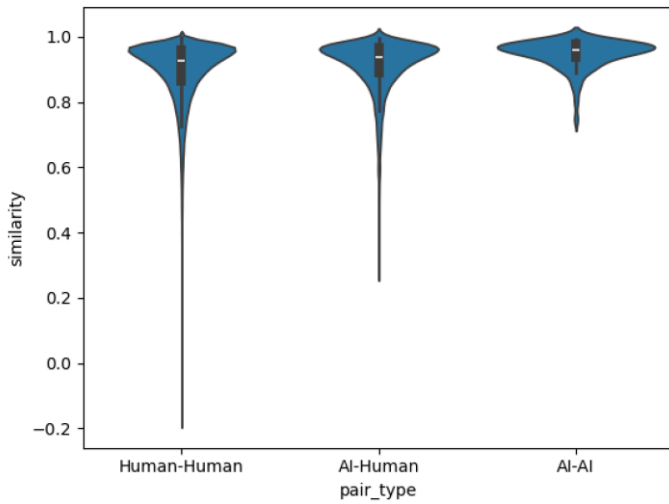
### Connection:

- Reproducibility
- Intellectual property

### Limitations:

- AI dataset size
- Number of audio features

# Visualizations



| cluster | 0    | 1    |
|---------|------|------|
| is_ai   |      |      |
| 0       | 4583 | 3384 |
| 1       | 85   | 163  |

# Expansion

## Abstract

We plan on investigating the relationship between human-made music and AI-generated music through their musical features. Musical features refer to MFCCs (Mel-Frequency Cepstral Coefficients, variation in frequency), BPM (beats per minute, tempo), and chroma (mapping keys onto values, tonal consistency). We want to find out if human-made music is more similar to AI-generated music than two average human-made songs are to each other. To answer this question, we have four different datasets in use, specifically top 4000 Spotify songs ranging from 1960s-now, a control dataset of human music to calculate the similarity score between the human and AI music, a subsection of the human datasets that are sampled pairs as a benchmark for the similarity score, and a dataset of about 250 AI-generated songs from Suno AI. In extracting the musical features from the Spotify dataset, we utilized Chosic, a music categorization database. We plan on incorporating these datasets into a website capable of calculating the similarity score between a pair of songs, creating our own dataset of similarity scores in order to investigate the relationships between the features of human-made and AI-generated music.

## Background

Since 2022, generative AI has become a hot topic in creative industries. AI art is all over Instagram, TikTok, and YouTube with videos, images, and music. There are several methods of identifying AI videos and images, since visual cues can be easily received. However, AI music has been especially challenging to detect, YouTube playlists rarely utilize the AI-generated label, experimenters are finding workarounds to publish on music streaming platforms, and major record labels are filing copyright suits. We want to investigate the musical features of AI and human music to compare their similarities and interpret how the industry will be affected. This question relates to musical information retrieval, which is a research field reliant on extracting features from audio files. Three frequently extracted features are Mel-Frequency Cepstral Coefficients (MFCCs), beats per minute (BPM), and chroma.

## Methods

Sampling

Datasets

Suno AI

Chosic

Paired samples

Lit Review

Feature Extraction

Exploratory data analysis

Cosine similarity

D Cohens

Violin plot

K-Means cluster, silhouette score

PCA

K-Means

Actual labels

Confusion Matrix

Results

Discussion

Limitations

Future Work

# Questions to Address

- What is the question you ask and why do we care?
- How is it connected to the class?
- What is the answer you found?
- What methods did you use to find the answer?
- What are the pitfalls you avoided?
- What are the limitations?

Site



Site QTA

State the question you asked succinctly.

- Why is the question important? Why do we care?
- What past work has been done in this area? How are you building on/adding to that work?
- State the methodology you used to answer the question (literature review, reproducing existing results, performed a new study, wrote software to test an idea, etc).
- What were the initial potential pitfalls of using that methodology? How did you avoid them?
- Did you work with data about humans?  
Did you involve human subjects in your study?

Answer the question and explain what you did to perform this responsibly.

- Explain the progress you made this quarter on the project. What was difficult and how did you solve it?
- What is the answer to the question?
- What are the caveats and limitations of your work? What might you do in the future with more time/more resources?